

An end-to-end application for all
construction needs

A PROJECT REPORT

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Under the Guidance of

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*in partial fulfillment of the requirements for the degree
of*

BACHELOR OF TECHNOLOGY
in
COMPUTER SCIENCE ENGINEERING
with specialization in Computer Networking



DEPARTMENT OF NETWORKING AND
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SRM INSTITUTE OF SCIENCE AND
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Authors

ABSTRACT

One of the most important pillars of economic development is the construction sector. Nevertheless, the industry frequently experiences inefficiencies that result in project delays, cost overruns, poor communication, and inconsistent documentation. This is explained by the disintegration of operations, in which several disjointed systems collaborate to plan, carry out, procure, manage labour, and communicate with clients.

These conventional approaches frequently lead to poor management and decreased productivity because they are unable to provide coordinated control or real-time insights. By combining all stages of construction into a single, intuitive application, this project offers a novel, cohesive digital platform intended to address these systemic inefficiencies.

Every step of the building process, from the initial planning and design to the last project handover, will be accessible and controllable through the platform.

Project scheduling, labour allocation, document control, equipment rental, resource tracking, procurement, and stakeholder communication are important modules. This end-to-end application aims for both operational excellence and digital transformation. The solution will support proactive decision-making, accountability, transparency, and sustainability in the construction industry by utilizing cloud technologies, mobile access, and analytics.

The application enables smooth collaboration, waste reduction, and on-time and cost-effective project delivery for all stakeholders, whether working on a residential building or a commercial complex.

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ABBREVIATIONS

Abbreviation	Full Form
UI	User Interface
UX	User Experience
API	Application Programming Interface
DB	Database
CRUD	Create, Read, Update, Delete
JWT	JSON Web Token
REST	Representational State Transfer
SRS	Software Requirements Specification
MVC	Model View Controller
CI/CD	Continuous Integration / Continuous Deployment
HTML	Hypertext Markup Language
CSS	Cascading Style Sheets
JS	JavaScript
DBMS	Database Management System
SQL	Structured Query Language
QA	Quality Assurance
QA Testing	Quality Assurance Testing
IDE	Integrated Development Environment
JSON	JavaScript Object Notation
MVC	Model-View-Controller
CDN	Content Delivery Network
AWS	Amazon Web Services
S3	Simple Storage Service

Abbreviation	Full Form
UXD	User Experience Design
PM	Project Manager
DevOps	Development and Operations
TDD	Test-Driven Development
SMTP	Simple Mail Transfer Protocol
HTTPS	Hypertext Transfer Protocol Secure
SSL	Secure Sockets Layer
CRUD	Create, Read, Update, Delete
OTP	One-Time Password
B2C	Business to Customer
P2P	Peer to Peer
UI/UX	User Interface / User Experience
CSV	Comma-Separated Values

CHAPTER 1

INTRODUCTION

1.1 Introduction to an end-to-end application for all construction needs:

Construction processes are by their very nature multidisciplinary. Civil engineering, architecture, electrical design, materials procurement, cost estimation, labour management, etc. all combine to make construction an interesting and challenging process. Coordination and alignment through and among these processes is where experienced project managers find a major challenge. Commonly, different departments or contractors would manage their workflows with their own software or even manual tools, creating silos of information, and thus inefficiencies.

Because decision-making is invariably made reactively, this means a lack of real-time or near real-time data; the problems are found only after the impact in the form of delay or loss has resulted. All of these can potentially throw off timelines, therefore increasing costs: Lack of communication between clients, engineers, and suppliers. In the absence of a single digital infrastructure, objectives will further diverge, documentation will be scattered, and duplications will arise due to lack of coordination. The construction sector, though, is gradually adopting digital solutions for these challenges, most of which are expensive and non-scalable. One size fits all-type applications solving issues like accounting or inventory.

Our solution bridges this gap by providing a comprehensive but modular platform that serves all functions of construction. Project size ranging from thousands to millions will be scalable and customizable according to the needs of small or large enterprises. From digital drawings to Gantt-based scheduling, live updates from the site, automated material requests, and stakeholder communication tools, the application will now act as the command center for construction. The ideas embedded in this form of tool will, thus, revamp the construction world into a fairer, traceable, and more effective setup.

1.2 Motivation

The motivation of this project has come from it being inspired by the research into small and large projects in construction with respect to the latest challenges. Most contractors are still using antiquate manual systems for carrying out procurement, site inspections, and labor assignment tasks. This, in effect, slows down operations and creates a lot of room for misconceptions and errors to manifest. A simple miscommunication on the material deliveries could shut down operations for several days at a stretch, costing time as well as money. Our inspiration is rooted in the solutions that speak practical scalability for all sizes of construction companies for such problems.

The interviews conducted with the site managers, engineers, and clients revealed many pain areas including: uncoordinated schedules, mismatched supply orders, delays due to labor, wrongly made documentation, and regular disputes over budget allocation. Though there are some digital tools presently available in the market, most of them are quite expensive, neither customizable nor focusing much on finance and procurement. Clearly, the market has needed a very affordable end-to-end solution, which would be able to cover the gap between high-tech systems and usability at field level.

Another thing was the motivation to see a digital ecosystem where every stakeholder from architect to mason would have been able to hold up-to-speed interactions, updates, and alignments of their work in real time. This then promises better project management with improved control of quality, reduced disputes, improved client satisfaction, and a more sustainable construction model. The project intends to democratize digital access for the construction industry laying a ground for smart building practices throughout the country.

1.3 Sustainable Development Goal of the Project

Our project directly aligns with Sustainable Development Goal 9: Industry, Innovation, and Infrastructure, which emphasizes the need for resilient infrastructure and inclusive, sustainable industrialization.

The construction sector is large and substantially affects the economy, environment, and society, and innovations in construction can change the game. With our application bringing together all construction processes digitally, the way infrastructure is planned, built, and managed is kept up to date and becomes more resource-efficient and performance-focused.

The project is indirectly supportive of SDG 11: Sustainable Cities and Communities, achieving low-impact environmental construction processes managing well and good urban development. It supports environmentally sustainable construction practices by minimizing material wastage, allowing for energy-efficient designs, and smart resource planning. It also helps with the optimization of land use, planning for green infrastructure, and implementing smarter zoning and community designs.

The social and economic sustainability aspects of the project are further offered by the platform. Offering the smaller players access to very sophisticated tools, it enables inclusive growth and employment. It establishes cooperation among diverse stakeholders, reduces corruption through transparency, and ensures integrity in the delivery of infrastructure projects. Therefore, this project is a major stepping stone in advancing targets across multiple SDGs, viewed through the lenses of technology, efficiency, and accountability.

1.4 Product Vision Statement

1.4.1. Audience:

Construction firms, individual contractors, project engineers and architects, real estate developers, government agencies, and end users engaged in infrastructure development are the key audiences for this system.

Each user type is given access to role-specific tools and dashboards that assist them in carrying out their work more easily. Suppliers, equipment renters, labor hiring agencies, and quality auditors partake in the projects through restricted-access portals, thus are policy users.

1.4.2. Needs:

Primary Needs:

- A centralized system to plan and track projects
- Resource allocation and monitoring of labor in real time
- Scheduling and interdependency of activities
- Budgeting and cost control
- Communication and sharing of documents

Secondary Needs:

- Invoice and vendor management
- Legal documentation and compliance checks
- GIS-enabled site maps
- Material analytics and cost evaluation
- User feedback and performance scoring

1.4.3. Products:

The application is delivered via two types of interfaces: a web portal, which is for centralized office management, and a mobile application to be used in field-level operations.

The platform consists of a number of key modules, including:

- A project dashboard with real-time KPIs
- Inventory management linked to suppliers
- Workforce allocation and attendance

- Equipment scheduling and maintenance alerts
- Budgeting and financial overview
- Reports and progress charting
- Alerts, messaging, and audit logging

1.4.4. Values:

The application is built around five core values:

- Transparency: All activities and changes are tracked, thereby minimizing disputes.
- Efficiency: The automated workflows eliminate delays and redundant work.
- Scalability: It supports anything from small housing projects through to multi-billion-dollar developments.
- Simplicity: A good UI/UX design that is friendly for both tech-savvy and non-technical users.
- Trust: With role-based access, data integrity and secure collaboration comes guaranteed.

1.5 Product Goal

To totally transform the management of construction projects into a new kind of central activity, it works on a complete, easy-to-use, and entirely scalable solution that meets every aspect of the workflow.

It suits everyone who might be a contractor building one house or a developer building a smart city, depending on their way and stage of operation.

It aims to become the most complete tool for no longer needing different software and several manual paperwork in workflow, now transforming it into a viable format.

Short-term: better communication, human error reduction, and on-time project delivery possible through automation and real-time tracking, stretching from site updates to procurement approvals using a single interface.

Long-term, however, the dream is to have this turn into an ecosystem for construction-a digital marketplace where clients will be able to hire professionals while materials would be listed by suppliers and tenders would be turned directly through the app by contractors.

Ultimately, the product has been envisioned as the backbone for smart building construction.

Using data analytics and AI integration in future versions, the product will eventually help to predict risks, optimize resource usage, and provide intelligence-driven recommendations. The more precise goal is not to build but to build smarter, faster, and more sustainably.

1.6 Product Backlog

Table 1.1 Detailed User Stories

S.No	User Stories of AI E-Learning Application
#US 1	As a new user, I want to sign up as a client, builder, or admin, so that I can assess the relevant features of the platform
#US 2	As a registered user, I want to log in securely, so that I can access my account and use the application.
#US 3	As an admin, I would like to verify and approve builder registrations so that only authenticated and reputed builders are listed.
#US 4	As a builder, I want to create and manage my profile information with company details, services, and past projects, so that clients could trust and choose me.
#US 5	As a client, I would like to view and browse among verified builders and construction companies to find a suitable service provider.
#US 6	As a client, I would like to view limited information on builders without logging in, so I can explore before signing up.
#US 7	As a client, I want to search and filter builders by location, services, and ratings, so that the choices will be relevant for me.
#US 8	As a client, I'd like to be able to contact the builder either using a contact form or any other contact details that would be shared, to communicate my construction needs.
#US 9	As a client, I want to view a detailed profile of builders upon login so that I can make an informed decision.
#US 10	As an Admin, I would like to be able to manage users, listings, and reported issues, keeping the platform clean, secure, and trustworthy.
#US 11	As a builder, I would like to see analytics (such as views, lead generation) to better understand my business performance on the platform.

The product backlog for "End-to-End Application for All Construction Needs" was set up on an MS Planner Agile Board, as presented in Figure 1.1. The Product Backlog has defined a full set of user stories covering the entire scope and functionality of the application.

Each user story consists of necessary parameters like:

MoSCoW prioritization, Functional and non-functional parameters, detailed acceptance criteria with linked tasks.

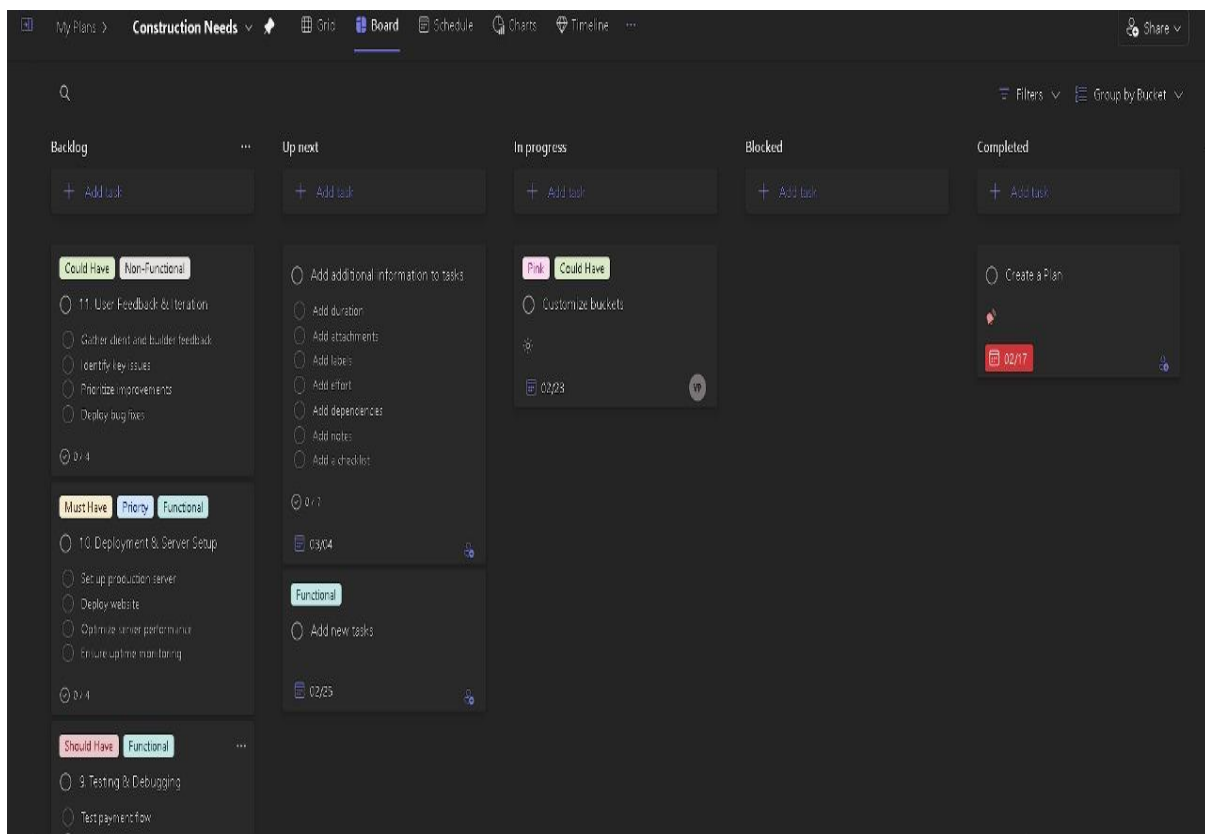


Figure 1.1 MS Planner Board of an end-to-end application for all construction needs

1.7 Product Release Plan

The following Figure 1.2 depicts the release plan of the project

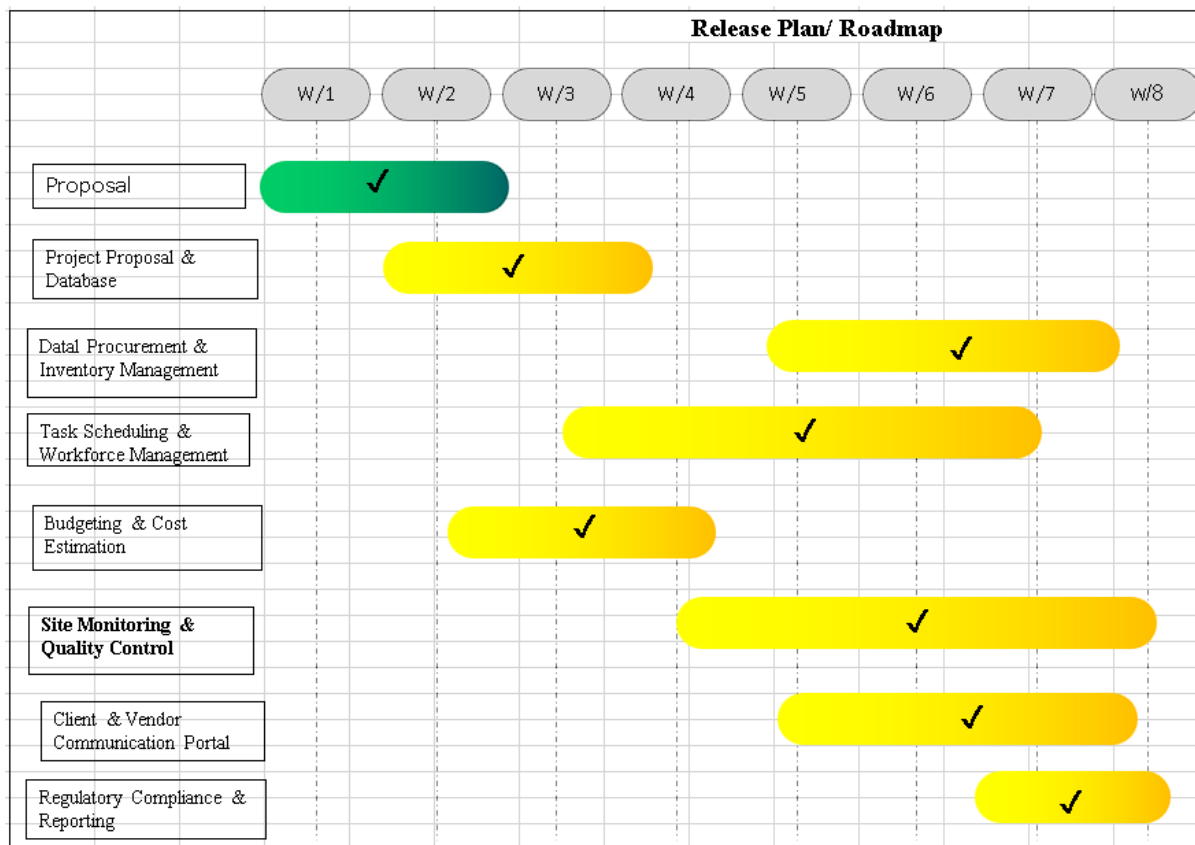


Figure 1.2 Release plan of an end-to-end application for all construction needs

CHAPTER 2

SPRINT PLANNING AND EXECUTION

2.1 Sprint 1

2.1.1 Sprint Goal with User Stories of Sprint 1

The Goal of the first sprint is to construct the user landing page and to enable the search functionalities such as skills and courses.

The following table 2.1 represents the detailed user stories of the sprint 1

Table 2.1 Detailed User Stories of sprint 1

S.NO	Detailed User Stories
US #1	As a new user, I want to sign up as a client, builder, or admin, so that I can assess the relevant features of the platform
US #2	As a registered user, I want to log in securely, so that I can access my account and use the application.
US #3	As an admin, I would like to verify and approve builder registrations so that only authenticated and reputed builders are listed.
#US 4	As a builder, I want to create and manage my profile information with company details, services, and past projects, so that clients could trust and choose me.
#US 5	As a client, I would like to view and browse among verified builders and construction companies to find a suitable service provider.

Planner Board representation of user stories that were addressed in Sprint-1 :

The image shows a detailed view of a Planner Board card. At the top, it's titled '1) User Authentication & Role Management' under the 'Construction Needs' group. Below the title is an 'Assign' button. There are two tags: 'Must Have' (yellow) and 'Priority' (blue). The card is organized into several sections: 'Bucket' (Backlog), 'Progress' (In progress), 'Priority' (Important), 'Start date' (02/01/2025), 'Due date' (02/15/2025), and 'Repeat' (Does not repeat). A 'Notes' section contains the text 'Implement secure login/signup for clients, builders, and admins with role-based access.' Below this is a 'Checklist 4 / 4' section with four items: 'Design authentication system', 'Implement login and signup', 'Set up role-based access', and 'Conduct security testing', all of which are checked. There is an 'Add an item' option at the bottom of the checklist. The 'Attachments' section shows a screenshot of a SharePoint page with the URL 'https://srmist.sharepoint.com/'.

Construction Needs

1) User Authentication & Role Management

Assign

Must Have Priority

Bucket: Backlog Progress: In progress Priority: Important

Start date: 02/01/2025 Due date: 02/15/2025 Repeat: Does not repeat

Notes

Implement secure login/signup for clients, builders, and admins with role-based access.

Checklist 4 / 4

- Design authentication system
- Implement login and signup
- Set up role-based access
- Conduct security testing

Add an item

Attachments

Screenshot 2025-04-20 22
<https://srmist.sharepoint.com/>

Figure 2.1 User story-1

Construction Needs

☐ 2) Builder Verification System

Assign

Should Have

Must Have

Functional

Bucket

Backlog

Progress

☐ Not started

Priority

Medium

Start date

02/10/2025

Due date

02/25/2025

Repeat

Does not repeat

Notes

☐ Show on card

- Develop a document submission and verification system for builders.

Checklist 4 / 4

☐ Show on card

☒ Define required documents

☒ Build submission interface

☒ Implement admin approval/rejection

☒ Set up notification system

☐ Add an item

Attachments

Screenshot 2025-04-20 22

https://srmist.sharepoint.com/

☒ Show on card

Add attachment

Figure 2.2 User story-2

3) Business Listings & Search

Completed on 04/01/2025 by CHIRANJEEV KUMAR (RA2211029010019)

Assign

Should Have

Bucket

Backlog

Progress

Completed

Priority

Medium

Start date

02/20/2025

Due date

03/10/2025

Repeat

Does not repeat

Notes

☐ Show on card

Create a search system to filter builders by location, expertise, and ratings.

Checklist 0 / 4

☒ Show on card

☐ Design database schema

☐ Implement search algorithm

☐ Develop UI for search results

☐ Test search functionality

☐ Add an item

Attachments

Add attachment

Comments

Type your message here

Figure 2.3 User story-3

Construction Needs

○ 4) Profile & Portfolio Management

Assign

Add label

Bucket

Backlog

Progress

○ Not started

Priority

● Medium

Start date

03/05/2025

Due date

03/20/2025

Repeat

↺ Does not repeat

Notes

Show on card

Allow builders to showcase past projects and services.

Checklist 4 / 4

Show on card

✓ Design builder profile template

✓ Implement profile editing feature

✓ Enable portfolio image uploads

✓ Set up validation checks

○ Add an item

Attachments

Screenshot 2025-04-20 22

https://srmist.sharepoint.com/

...

✓ Show on card

Add attachment

Figure 2.4 User story-4

14

Construction Needs

✓

5. Client Requests & Messaging

Completed on yesterday by you

Assign

Add label

Bucket

Backlog

▼

Progress

✓ Completed

▼

Priority

● Medium

▼

Start date

03/15/2025

Due date

04/05/2025

Repeat

↺ Does not repeat

▼

Notes

☐ Show on card

: Enable clients to send service requests and communicate with builders

Checklist 1 / 4

✓

 Show on card

✓

 Create request submission form

Implement message system

Ensure real-time updates

Test UI for messaging

Add an item

Attachments

Add attachment

Comments

Figure 2.5 User story-5

2.1.2 Functional Document

2.1.2.1. Introduction

The End-to-End Application for All Construction Needs attempts to create a centralized digital platform that binds the client to the verified construction service provider. The platform enables more efficient discoverability, verification, and hiring of reputed builders and construction companies. The platform stands for trust, transparency, and convenience for various users, all with construction-related requirements.

2.1.2.2. Product Goal

The primary goal of this project is to simplify and digitize the quest for construction services, creating a verified ecosystem of service providers and smart access to information. The platform will, therefore:

- Connect clients to verified and reputed builders or construction companies.
- Render limited access into the builder info for unregistered users to encourage sign-ups.
- Allow builders to showcase their services and performance metrics.
- Provide a centralized admin dashboard for user management and platform oversight.

2.1.2.3. Demography (Users, Location)

Users:

- Target Users: Individuals or companies looking for construction services (clients), verified builders and construction firms (service providers), and Platform Admins.
- User Characteristics: High-tech and semi-tech users who are actual decision-makers in a residential or commercial construction project, business-oriented concentrating on finding builders.

Location:

- Target Location: Urban and semi-urban areas where there is active real estate development, especially in the circumstance where demand for verified, quality-driven construction services exist.

2.1.2.4. Business Processes

User Registration and Authentication:

- Users register as client, builder, or admin using their email addresses.
- The authentication process allows secure access and use of role-specific features.
- Builders should be verified by administrators before they appear on the list.

Service Listing and Builder Profile Management:

- Builders can create a comprehensive profile with information on service areas, past work, and contact information.
- Clients may be able to see basic information without logging in and get full access only after signing up.

Admin Moderation and Verification:

- Admins moderate activity on the platform, verify builder credentials, and handle complaints or reports of listings.

2.1.2.5. Features

This project focuses on implementing the following key features:

Feature 1: User Registration

1. Description:

- The platform has a secure, role-based registration and login for clients, builders, and admins.

2. User Story:

- As a user, I want to register according to my role and log in to use all features of the platform that are relevant to me.

Feature 2: Builder Profile Management

1. Description:

- Verified builders may create, edit, and manage profiles that portray their services, experience, and credentials.

2. User Story:

- As a builder, I want to showcase my services and past projects to attract potential clients.

Feature 3: Enhanced Search Functionality

1. Description:

- Clients can view basic information on builders using guest access whereas full profile content is accessible to registered clients only.

2. User Story:

- As a client, I want to know some specifics about the builder before deciding whether or not to register on the platform.

2.1.2.6. Authorization Matrix

Table 2.2 Access level Authorization Matrix for Sprint-1

Role	Access Level
Administrator	Full access to user management, builder verification, complaint handling, and platform settings.
Builder	Access to profile creation, project uploads, service listings, and performance insights.
Client	Access to search builders, view full profiles post-login, and contact service providers.
Guest User	Limited access to view basic builder information and explore platform features.

2.1.2.7. Assumptions

- Before going live, builders registering on the platform will submit relevant documents for the admin's verification.
- Clients shall be allowed limited-to-no browsing before sign-up in order to promote engagement on the platform.
- The development team shall have timely continuous access to cloud infrastructure and a sandbox environment for testing.
- All platform users shall provide feedback during testing and deployment and aid the iterative development process.
- The application will comply with data privacy and security regulations to protect user confidentiality.
- The platform shall effectively scale to hold an expanding user base comprising builders and clients from different corners of the globe.

2.1.3 Architecture Document

2.1.3.1. Application

Microservices-based Architecture:

The platform is based on a modular microservices architecture for independent development and scaling of each core functionality. Some of the key services are:

- **Authentication Service:** Provides user registration, log-in, password encryption, and secure token-based authentication for all roles (client, builder, admin).
- **User Role Management Service:** Manages access privileges and role-based dashboards to clients, builders, and admins.
- **Builder Management Services:** Enables builders to create/update their profiles, list services they provide, and upload portfolio projects. Admins use this service for builder verification.
- **Client Search & Filter Service:** Allows searching and filtering the builder profiles by parameters such as location, service types, ratings, etc.
- **Admin Dashboard Service:** Provides features for user moderation, verification workflows, issue tracking, and analytics.
- **Communication & Contact Service:** Handles client inquiries and forms for builders to communicate seamlessly.

2.1.3.2 System Architecture-

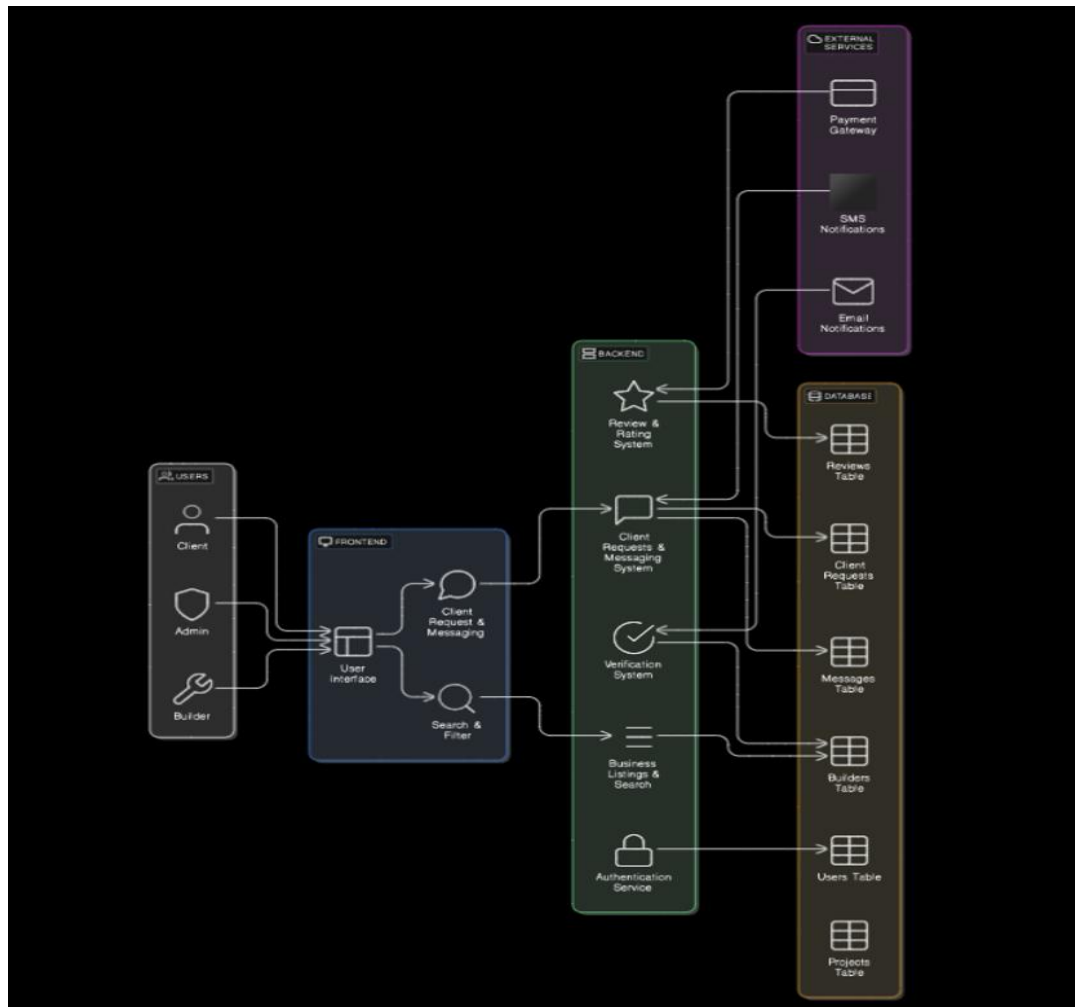


Figure 2.6 System Architecture Diagram

2.1.3.3. Data Exchange Contract:

Frequency of Data Exchanges:

Data exchanges are managed with careful consideration of timing and performance:

- **Real-Time Exchanges:** Critical operations such as **user login**, **builder verification status**, and **client-builder interactions** are handled in real-time via REST APIs to ensure instant updates and responsiveness.
- **Periodic Syncs:** Non-critical data like **site visit analytics**, **client browsing history**, and **builder activity logs** are synchronized at scheduled intervals (e.g., hourly or daily) to optimize system performance.

Data Sets:

The different data sets are managed on the platform, each with its data exchange requirements:

- **User Data:** It consists of usernames, passwords, contact details, and roles for users: admin, builder, or client. Such data can be exchanged whenever a user registers himself, logs in, or modifies any aspect of their profile.
- **Builder Data:** It involves the listings of provided services, company profiles, portfolio pictures, and verification status. It may be updated anytime a builder modifies his profile, uploads new projects, or has the information approved by admins.
- **Client Interaction Data:** It monitors clients searching for potential builders, their submission of a contact form, and saving target builder profiles. Such builds are updated based on user browsing inquiries and activities.
- **Admin & Complaint Data:** It covers the complaint logs, user reports, and status of resolution, primarily exchanged when admins commence action or intervene.

Mode of Exchanges (API, File, Queue, etc.) :

Various methods are used for data exchange across the platform:

- **API (RESTful):** Used for **real-time communication** between frontend and backend services, such as user authentication, builder profile updates, and contact requests.
- **File-Based Exchanges (S3):** Utilized for **uploading builder portfolios**, company documents, and handling any bulk data import/export functionality.

2.1.4 UI DESIGN

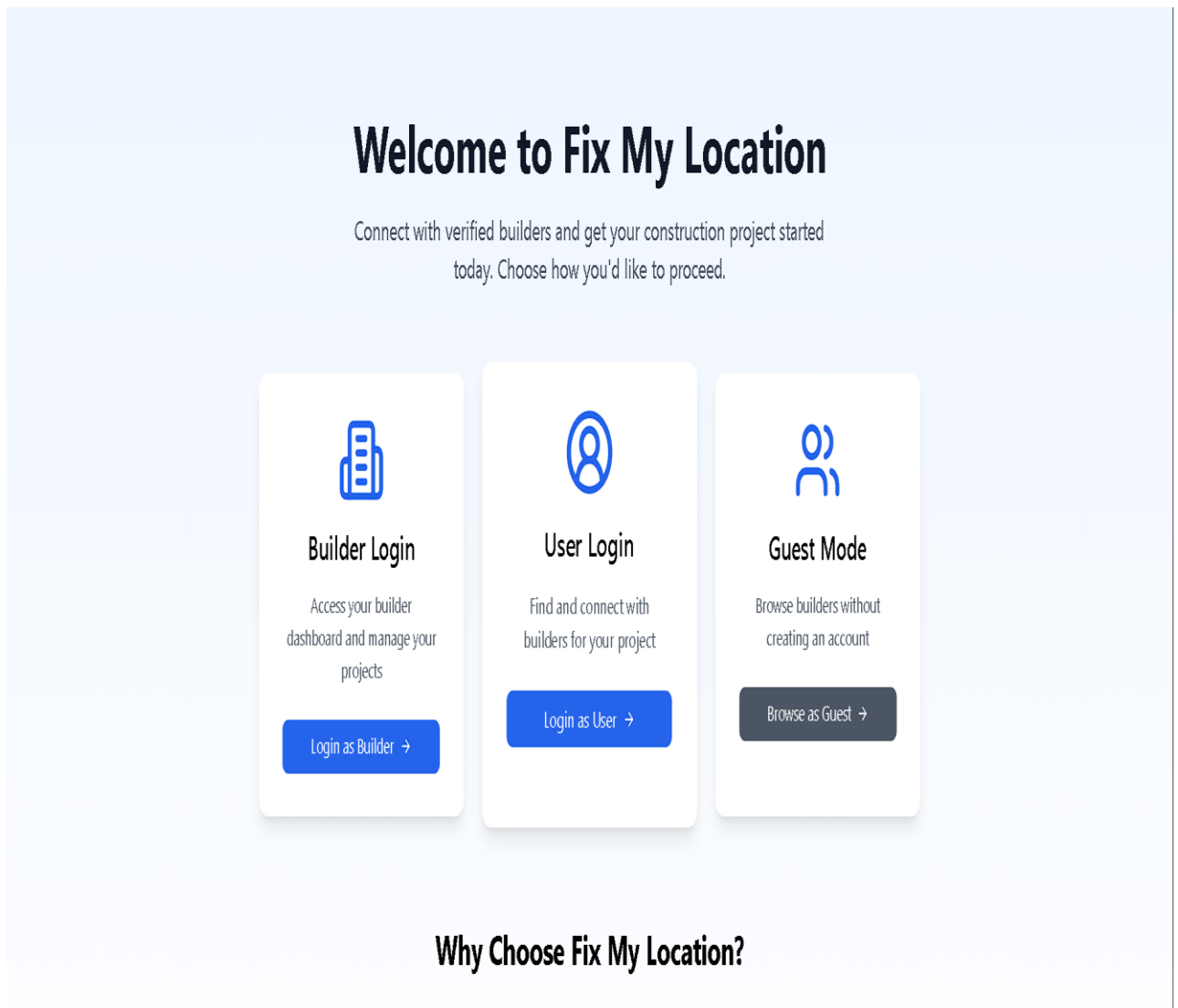


Figure 2.7 UI Design for Landing page



User Login

Find and connect with verified builders

Email Address



Password



Login

Demo credentials:

Email: user@example.com

Password: password

Figure 2.8 UI design for User login page

2.1.5 Schema Diagram

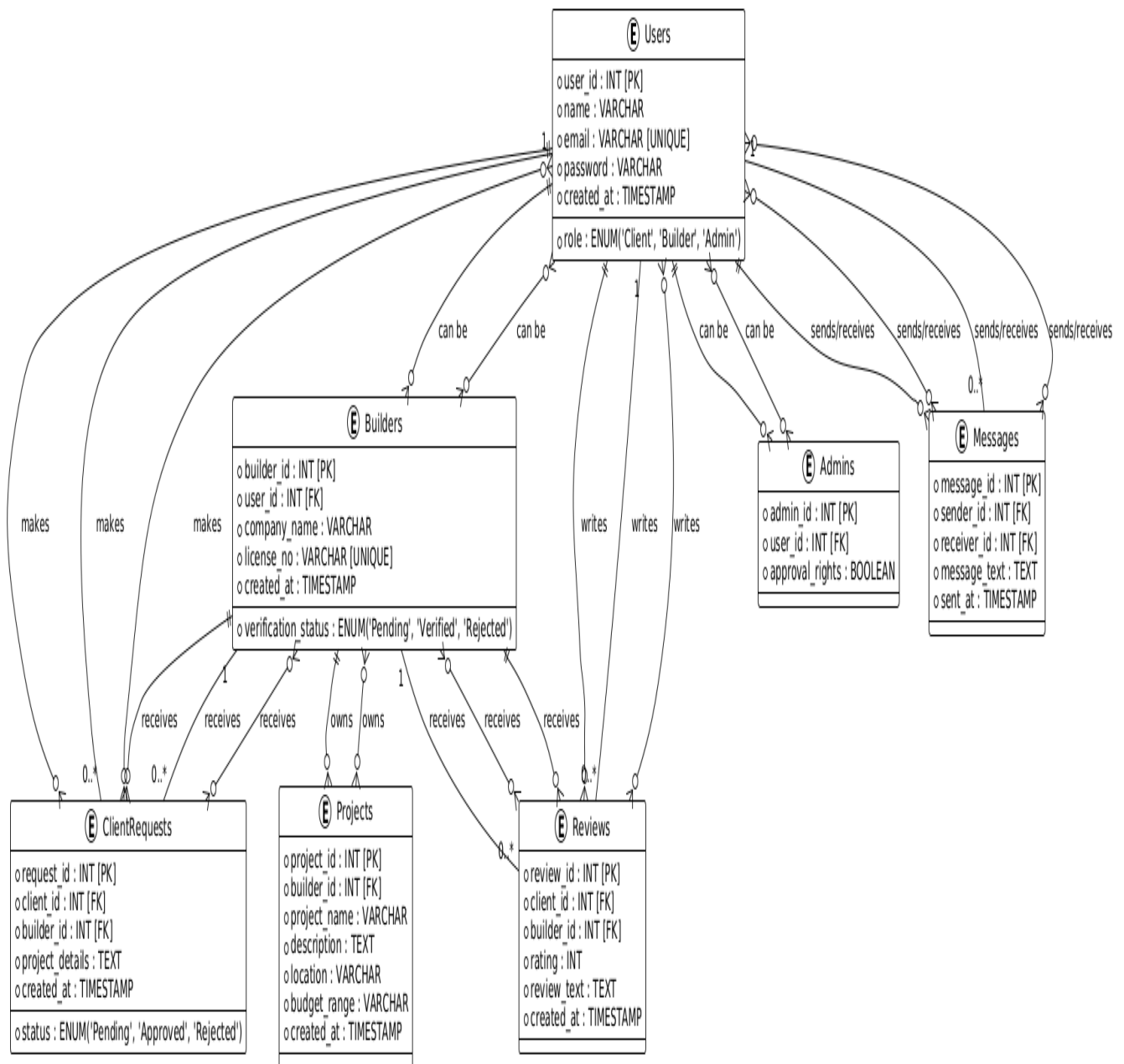


Figure 2.9 Schema Diagram of an end-to-end application for all construction needs

2.1.6 Daily Call Progress

Date	Team Members	Duration	Discussion Points	Tasks Assigned	Next Action Plan
10-Mar-2025	Chiranjeev, Vedant	1 hr	Brainstormed project idea and feasibility	Research on similar platforms	Finalize project scope and objectives
11-Mar-2025	Chiranjeev, Vedant	1.5 hrs	Identified core user problems	List user requirements	Prepare functional specs draft
12-Mar-2025	Chiranjeev, Vedant	1 hr	Drafted feature list for MVP	Group features into modules	Create wireframes
13-Mar-2025	Chiranjeev, Vedant	2 hrs	Reviewed wireframes and took feedback	Refine wireframes	Setup basic GitHub repo & folder structure
14-Mar-2025	Chiranjeev, Vedant	1 hr	Finalized tech stack options	Compare frontend/backend stacks	Freeze stack and environment setup
15-Mar-2025	Chiranjeev, Vedant	1.5 hrs	Setup project repo and dev environment	Install dependencies	Create initial folder & file templates
16-Mar-2025	Chiranjeev, Vedant	2 hrs	Discussed database schema	Draft initial ER diagram	Implement database models
17-Mar-2025	Chiranjeev, Vedant	1 hr	Talked about authentication strategy	Plan login/signup flows	Begin backend API for auth
18-Mar-2025	Chiranjeev, Vedant	1.5 hrs	UI discussion for login, dashboard	Sketch low-fi UI screens	Start frontend login screen
19-Mar-2025	Chiranjeev, Vedant	2 hrs	Finalized dashboard layout	Backend API endpoints mapped	Start integrating frontend-backend
20-Mar-2025	Chiranjeev, Vedant	1 hr	Discussed project structure improvements	Refactor some modules	Optimize routing and folder structure

Figure 2.10 Sprint-1 Standup meetings

2.1.7 Committed Vs Completed User Stories

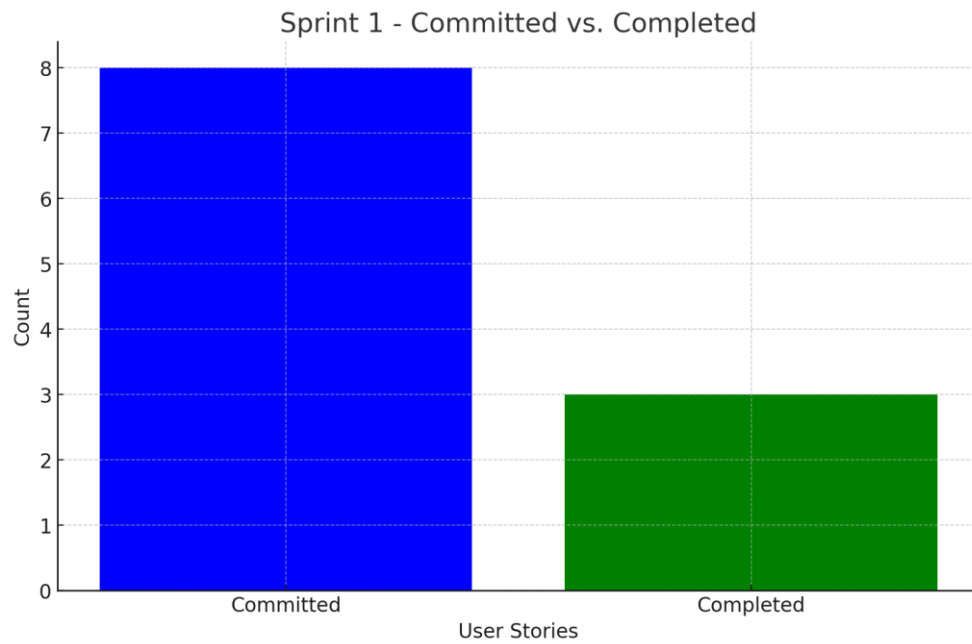


Figure 2.11 Sprint-1 Committed Vs Completed User Stories bar graph

2.2 SPRINT 2

2.2.1 Sprint Goal with User Stories of Sprint 2

The Goal of the second sprint is to implement the core client-facing features and refine the builder verification workflow, ensuring smooth interaction and data visibility between clients and builders.

The following table 2.3 represents the detailed user stories of the sprint 2

Table 2.3 Detailed User Stories of sprint 2

S.NO	Detailed User Stories
#US 6	As a client, I would like to view limited information on builders without logging in, so I can explore before signing up.
#US 7	As a client, I want to search and filter builders by location, services, and ratings, so that the choices will be relevant for me.
#US 8	As a client, I'd like to be able to contact the builder either using a contact form or any other contact details that would be shared, to communicate my construction needs.
#US 9	As a client, I want to view a detailed profile of builders upon login so that I can make an informed decision.
#US 10	As an Admin, I would like to be able to manage users, listings, and reported issues, keeping the platform clean, secure, and trustworthy.
#US 11	As a builder, I would like to see analytics (such as views, lead generation) to better understand my business performance on the platform.

Planner Board representation of user stories that were addressed in Sprint-2 :

The image shows a detailed view of a Planner Board card. At the top, it is titled '6. Review & Rating System' under the category 'Construction Needs'. Below the title are three action buttons: 'Assign', 'Add label', and 'Add label'. The card is organized into several sections: 'Bucket' (Backlog), 'Progress' (Not started), 'Priority' (Medium), 'Start date' (Start anytime), 'Due date' (04/10/2025), and 'Repeat' (Does not repeat). There is a 'Notes' section with a text area and a 'Show on card' checkbox. Below that is a 'Checklist 0 / 4' section with five items: 'Develop rating functionality', 'Create review submission system', 'Implement admin moderation', 'Test for fake review prevention', and 'Add an item'. At the bottom is an 'Attachments' section with one attachment: 'Screenshot 2025-04-20 22' with a URL 'https://smist.sharepoint.com/'.

Construction Needs

6. Review & Rating System

Assign

Add label

Bucket

Backlog

Progress

Not started

Priority

Medium

Start date

Start anytime

Due date

04/10/2025

Repeat

Does not repeat

Notes

Show on card

Type a description or add notes here

Checklist 0 / 4

Show on card

- Develop rating functionality
- Create review submission system
- Implement admin moderation
- Test for fake review prevention
- Add an item

Attachments

Screenshot 2025-04-20 22

https://smist.sharepoint.com/

Show on card

Figure 2.12 User story-6

...

×

Construction Needs

○ 7. Admin Dashboard & Security

Assign

Add label

Bucket

Backlog

▼

Progress

○ Not started

▼

Priority

● Medium

▼

Start date

Start anytime

Due date

04/20/2025

Repeat

↺ Does not repeat

▼

Notes

☐ Show on card

- Provide admins with controls for monitoring and managing users.

Checklist 0 / 4

☐ Show on card

○ Develop admin dashboard UI

○ Implement user activity logs

○ Add moderation tools

○ Conduct security testing

○ Add an item

Attachments

Screenshot 2025-04-20 22

https://srmist.sharepoint.com/

...

☒ Show on card

Figure 2.13 User story-7

...

×

Construction Needs

☐ 8. Payment & Subscription Model

Bucket

Backlog

▼

Progress

☐ Not started

▼

Priority

● Medium

▼

Start date

Start anytime

Due date

Due anytime

Repeat

Does not repeat

▼

Notes

☐ Show on card

Type a description or add notes here

Checklist 2 / 4

☐ Show on card

☒ Design subscription model

☐ Integrate payment gateway

☒ Implement user access control

☐ Test payment flow

☐ Add an item

Attachments

Screenshot 2025-04-20 22

https://srmist.sharepoint.com/

...

☒ Show on card

Figure 2.14 User story-8

...

×

Construction Needs

○

9. Testing & Debugging

Assign

✎

Should Have ✕

Functional ✕

Bucket

Backlog

▼

Progress

○

Not started

▼

Priority

●

Medium

▼

Start date

Start anytime

📅

Due date

Due anytime

📅

Repeat

🔄

Does not repeat

▼

Notes

☐ Show on card

- Ensure all functionalities work smoothly before deployment.

Checklist 0 / 4

☐ Show on card

○

Test payment flow

○

Conduct integration testing

○

Fix critical bugs

○

Final security audit

○

Add an item

Attachments

🖼️

Screenshot 2025-04-20 22

https://smist.sharepoint.com/

...

☒ Show on card

Figure 2.15 User story-9

...

×

Construction Needs

○

10. Deployment & Server Setup

Assign

✕

Must Have

✕

Priority

✕

Functional

✕

Bucket

Backlog

▼

Progress

○ Not started

▼

Priority

● Medium

▼

Start date

Start anytime

📅

Due date

Due anytime

📅

Repeat

🔄 Does not repeat

▼

Notes

☐ Show on card

Deploy the website and ensure server stability.

Checklist 0 / 4

☐ Show on card

○ Set up production server

○ Deploy website

○ Optimize server performance

○ Ensure uptime monitoring

○ Add an item

Attachments

 Screenshot 2025-04-20 22
<https://srmist.sharepoint.com/> ...

☒ Show on card

Figure 2.16 User story-10

...

×

Construction Needs

○ 11. User Feedback & Iteration

Assign

Could Have

×

Non-Functional

×

Bucket

Backlog

▼

Progress

○ Not started

▼

Priority

● Medium

▼

Start date

Start anytime

Due date

Due anytime

Repeat

↺ Does not repeat

▼

Notes

☐ Show on card

Collect user feedback and make necessary improvements

Checklist 2 / 4

☐ Show on card

✓ Gather client and builder feedback

○ Identify key issues

✓ Prioritize improvements

○ Deploy bug fixes

○ Add an item

Attachments

Screenshot 2025-04-20 22

https://srmist.sharepoint.com/

...

☒ Show on card

Figure 2.17 User story-11

2.2.2 Functional Document

2.2.2.1 Introduction:

In Sprint 2, the main focus of the End to End Application for All Construction Needs is building an initial core client-facing functionality and building up the verification pipeline of the builders. In the same vein, this sprint aims to enhance discoverability :- verified builders will now have detailed profiles available to logged-in users, and the builder verification system would become fully operational. This will shape the foundation for any and all transparent and efficient interaction between clients and construction services.

2.2.2.2 Product Goal

The objective of Sprint 2 is to conduct implementation and validation of key user flows that influence the usability and trustworthiness of the platform. It includes:

- Enabling clients to search for and view verified builders based on service type and location.
- Allowing only logged-in users to access complete builder profiles to promote registrations.
- A functional workflow for builder verification, whereby builders submit their needed documents and the admin approves or denies their verification status.
- A secure contact form that enables client-to-builder communication.

It provides feature characteristics for trust and drives engagement while ensuring builder visibility on the platform.

2.2.2.3 Demography (Users, Location)

Users:

- Target Users: Homeowners, landowners, real estate investors, property developers, etc., and construction service providers such as builders and contractors.
- User Characteristics: Users can range in their levels of technical proficiency, from tech-savvy professionals to non-technical users simply looking for trusted construction services. Their immediate need is therefore ease of access to verified professionals and transparent service-related information.

Location:

- Target Location : Initially targeting developments in urban and semi-urban areas with rapidly increasing internet opportunities, as well as a good culture for e-learning and collaboration

2.2.2.4 Business Processes

The core business processes put into effect in Sprint 2 include:

1. Builder Search and Discoverability:

- Clients can search for construction companies or individual builders filtered based on location, service type (such as residential or commercial), and customer ratings.
- This simplified UI works its way through the search results.

2. Profile Verification Access to Builders:

- The users must register and log in to access a complete view of a builder's particulars, including certifications, reviews, previous projects, or a chat.
- Non-registered guests can only see a small amount of profile information, enticing them to sign up.

3. Verification Workflow for Builders:

- Builders can upload any required documentation (including the license, project proofs, and identity validation).
- An admin dashboard would enable the platform admins to approve or reject build-verification requests based on review criteria.

4. Client-to-Builder Communication:

Once logged in, the client can contact a verified builder using the integrated contact form or messaging module for inquiries, quotes, or cooperation.

2.2.2.5 Features

The following prominent features were developed and integrated as part of Sprint 2, which were aimed to improve usability of the platform, increase the discoverability of builders, and heighten the engagement of users:

Feature 1: Search and Filter Builders

1. Description:

- Clients can search for builders using various filters such as location, service category, and user ratings.

2. User Story:

- As a client, I want to search and filter builders based on my requirements so that I can find suitable service providers quickly.

Feature 2: Detailed Builder Profile (Post Login)

1. Description:

- Once logged in, clients can view complete builder profiles, including company information, past projects, certifications, and customer reviews.

2. User Story:

- As a client, I want to view detailed builder profiles so that I can make an informed hiring decision.

Feature 3: Builder Document Upload for Verification

1. Description:

- Builders can upload documents such as licenses, work history, and ID proofs to get verified on the platform.

2. User Story:

- As a builder, I want to upload my documents so that I can be verified and build client trust.

Feature 4: Admin Verification Dashboard

1. Description:

- Admins can review submitted documents and either approve or reject builder verification requests through a centralized interface.

2. User Story:

- As an admin, I want to manage and verify builder profiles so that only credible service providers are visible to clients.

Feature 5: Client-to-Builder Contact Form

1. Description:

- A secure and simple contact form allows clients to reach out to builders with project inquiries or requests.

2. User Story:

- As a client, I want to contact a builder directly through the platform so that I can discuss my construction requirements.

2.2.2.6 Authorization Matrix

Table 2.3: Access Level Authorization Matrix for Sprint-2

Role	Access Level
Administrator	Full access to user management, builder verification workflows, and overall platform settings.
Builder	Access to profile creation, service listing, document uploads for verification, and client communication.
Client/User	Access to search and view builder profiles, contact verified builders, and manage their own user dashboard.
Guest User	Limited access to browse builder listings and view basic public information without login.

Architecture Document

2.2.3.1 Application

Microservices :

The application is architected modularly into microservices-form, i.e., core platform functionalities embedded in every service so that all modules can be independently developed, scaled, and maintained. Built and integrated in Sprint 2 are active ones, such as:

- **Builder Profile Management Service:** Enables manage create, update, and send detailed builder profiles, including history about projects, certifications, and service details.
- **Document Verification Service:** Manages secure document upload from builders and provides an interface on admin side to check the status of the documents and provide feedback.
- **Search and Filtering Service:** Clients can search for builders dynamically via attributes such as location, service type, and user rating.
- **Client-to-builder communication service:** Facilitates clients and builders to message each other, via a secure form, without revealing any contact details between them.

2.2.3.2 Data Exchange Contract

Frequency of Data Exchanges:

Data Exchanges are designed based on the reactive, secure, and operationally efficient principles:

- **Real-Time Exchanges:** These are critical to operations for user login, builder viewing profile, document uploads, and contact form submissions. This would be instantaneous through REST APIs so that there is no interruption to user experience.
- **Periodic Syncs:** Non-critical information such as user activity logs, updates to the status of the builders, or admin analytics would be synchronized from one system to another over an established time frame that would optimize system performance.

Data Sets:

The application handles and exchanges many data sets with corresponding rules and trigger points for each of them.

- **User Data:** This includes any personally identifiable information, along with authentication credentialing, and user role allocations. It is shared when logging in, registering, and updating the dashboard.
- **Builder Data:** Builder profile data, service offerings, the documents for verification, and project portfolios are included. These data get exchanged during profile creation, verification processes, and search functions.
- **Admin Logs and Verification Data:** Logs the verification workflows, approvals for builders, validations for documents, and audit logs for internal purposes.
- **Search Metadata:** Capturing tags for the location, filters applied, and results for builders for analytics and input for user preferences.

Modes of exchanges and Deployment:

To facilitate communication between services and the front end, multiple exchange modes have been utilized.

2.2.4 UI Design

 **Fix My Location**

Builder Login [Login / Sign Up](#)



Builder Login

Access your builder dashboard

Email Address



Password



Login

Demo credentials:
Email: builder@example.com
Password: password

Figure 2.18 UI Design for Builder Login page

Fix My Location

Connect with verified builders and get your project started today

 Search services...

 Location

 Any Rating

 Any Experience



 GST Verified

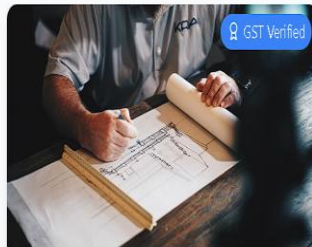
R K CONSTRUCTIONS

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2nd Floor, No. 1/2, Ramcons Fortuna Towers,
 * Kodambakkam high road, Nungambakkam,
 Chennai, Chennai, Tamil Nadu, 600034

 Projects Completed: 200

Figure 2.19 UI Design for Home Page

2.2.5 Functional Test Cases

Functional Test Case Template - Builders Portal						
Feature	Test Case	Steps to execute test case	Expected Output	Actual Output	Status	More Information
User Registration	Verify new user sign-up	Open website → Click "Sign Up" → Enter details → Submit	User should be successfully registered and redirected to the dashboard	The user is successfully registered. The application redirects the user to the home page.	Passed	Ensure required fields are validated
Builder Verification	Verify builder document upload	Log in as builder → Upload verification documents → Submit	Documents should be uploaded and under review	The documents are uploaded, and a verification request is generated. The system updates the builder's status to "Under Review."	Passed	Check error handling for invalid files
Search for Builders	Validate builder search functionality	Enter builder name in search bar → Click "Search"	Builders matching the search should be displayed	The system displays a list of builders matching the search criteria. The filters apply correctly, and pagination works as expected.	Passed	Test with different filters
Service Request Creation	Verify clients can request services	Log in as client → Select service category → Submit request	Request should be successfully submitted	The request is created but is not visible in the builder's dashboard. The system does not send a notification to the builder.	In Progress	Issue with request submission
Messaging System	Verify client-builder messaging	Open message section → Send a message to a builder → Check for response	Message should be delivered and visible to recipient	The message is sent but does not appear in the conversation immediately. The recipient receives a delayed notification.	In Progress	Messages sometimes delayed

Table 2: Functional Test Cases

2.2.6 Daily Call Progress

Date	Team Members Present	Duration	Discussion Points	Tasks Assigned	Next Action Plan
10-Apr-2025	Chiranjeev, Vedant	1 hr	Finalized features for MVP and user roles	Draft use-case diagram	Start system architecture design
11-Apr-2025	Chiranjeev, Vedant	1.5 hrs	Designed system architecture (5-layer model)	Document architecture, select tech stack	Create PlantUML diagram and ER model
12-Apr-2025	Chiranjeev, Vedant	2 hrs	Worked on user management module (login, signup, roles)	Chiranjeev: Backend APIs, Vedant: UI screens	Test API endpoints
13-Apr-2025	Chiranjeev, Vedant	1 hr	Integrated frontend with backend for login and dashboard	Fix UI issues, update error handling	Move to project management module
14-Apr-2025	Chiranjeev, Vedant	1.5 hrs	Discussed project tracking and Gantt chart integration	Research chart libraries, mock design	Begin implementing progress tracker feature

Figure 2.20 Sprint-2 Standup meetings

2.2.7 COMMITTED Vs COMPLETED USER STORIES

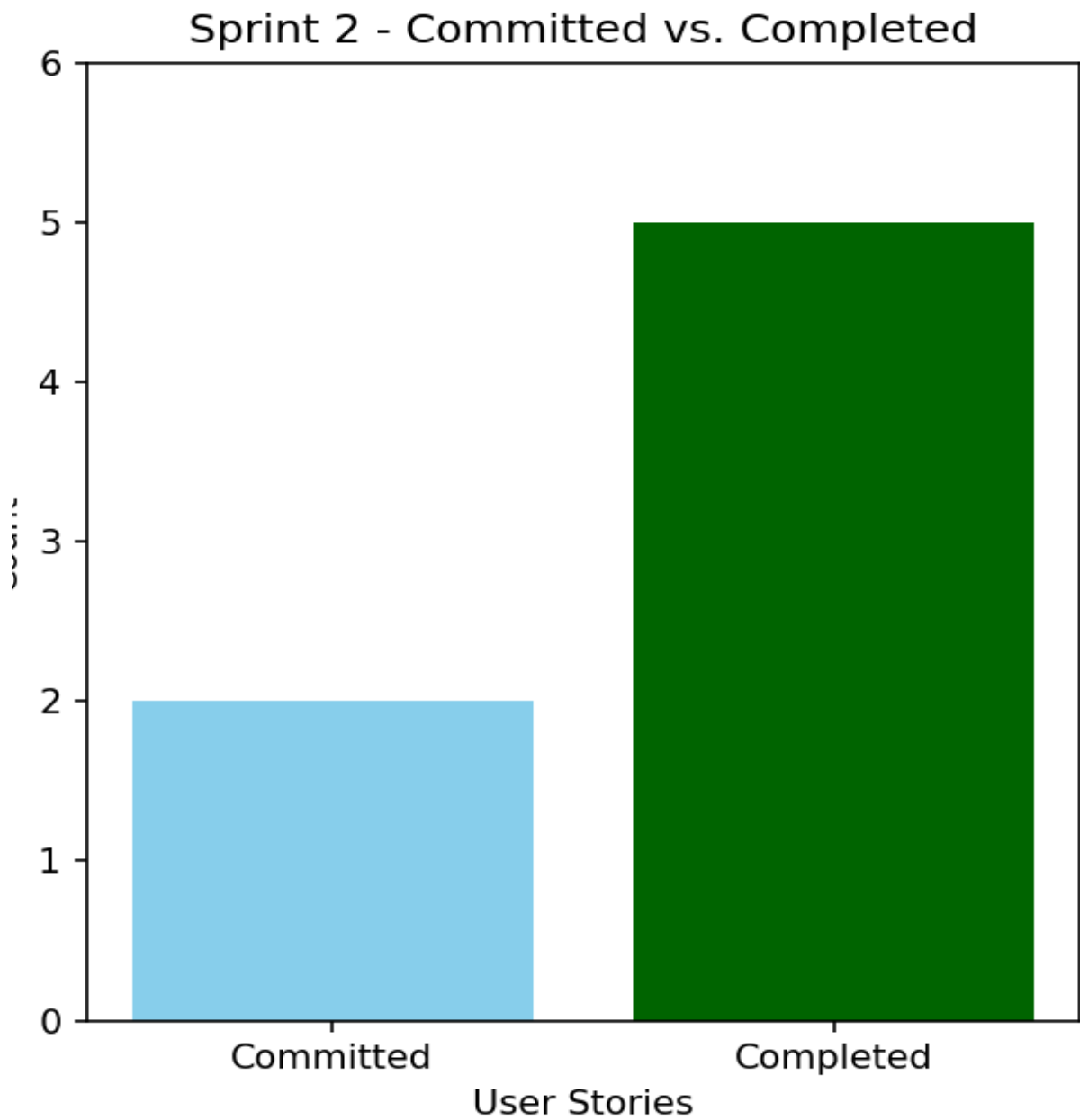


Figure 2.21 Sprint-2 Committed Vs Completed User Stories Graph

2.2.8 Sprint Retrospective

Sprint Retrospective-Builders Portal			
What went well	What went poorly	What ideas do you have	How should we take action
All assigned tasks were completed on time. Team members adhered to the sprint deadlines.	Requirements changed mid-sprint. Some features had to be modified at the last minute.	Plan for a buffer to handle scope changes. Introduce a flexible sprint planning approach.	Set stricter deadlines for finalizing requirements. Establish a clear requirement freeze deadline.
Collaboration and communication were seamless. Daily stand-ups were effective in keeping everyone aligned.	API response times were slow. Performance issues caused delays in data retrieval.	Optimize API calls and caching mechanisms. Use load testing tools to monitor performance.	Improve server performance by implementing caching. Use database indexing and optimize queries.
The bug resolution process was efficient. Critical issues were identified and fixed before the release.	Testing coverage was inadequate. Some edge cases were missed, leading to post-release bugs.	Improve test coverage with automated testing. Implement unit and integration testing for key functionalities.	Create a dedicated testing phase before deployment. Include QA reviews and automate regression tests.
User feedback from the prototype was positive. Clients found the UI intuitive and easy to navigate.	Task dependencies caused bottlenecks. Some features were delayed due to waiting on prior implementations.	Introduce a dependency tracking system. Ensure tasks with dependencies are prioritized correctly.	Use project management tools to track dependencies. Assign dependencies early and set up reminders.

Table 2.6 Sprint Retrospective

CHAPTER 3

RESULTS AND DISCUSSION

3.1 Project Outcomes

The end-to-end construction service platform's design and development yielded several key outcomes with respect to the initial objectives and user requirements. The bridging of the gap between clients and verified builders is established through a reliable and accessible digital platform. Major results that transpired all through the project are listed below:

- A Live Website:
 - All aspects of the platform design and development were carried out such that the live website offers a fully functional service as a responsive web interface.
 - Users are able to easily browse through a catalogue of construction service providers, view verified builder profiles, and access service requirements.
- User Role and Verification:
 - The system is designed to segregate three types of users-clients, builder, and admin. There is a verification process for builders before the activation of their profile, which ensures the credibility and trustworthiness of the listed service providers.
 - The admins operate the platform; monitor the listings and verifications.
- User Authentication and Data Privacy
 - The user authentication for all roles is secured by password encryption, session management, and role-based access control.
 - Data integrity and privacy were given crucial consideration along the whole development process.
- Easy and Efficient Service Discovery and Navigation
 - The user interface supports easy navigation and service discovery through filters, search, and categories, thereby granting user-friendliness and intuitiveness to the platform.

- Scalability for Future Integration
 - The architecture in consideration could also support future integrations for additional features such as chat support, payment gateways, project tracking dashboards, and feedback systems, thus allowing the platform to evolve with the needs of the users.
- Positive Feedback During Sprint Reviews
 - Building on feedback acquired, constructive remarks were made by the team across three Agile sprints, which iteratively improved the platform.
 - The majority of tasks committed to were effectively performed with increasing efficiency and clarity of the tasks in the latter sprints.
- Deployment and Testing
 - The platform has undergone extensive testing in terms of functionality, responsiveness, and usability. Feedback from trial users has been specifically incorporated to ensure that the system meets the requirements of usability.

3.2 Committed Vs Completed User stories

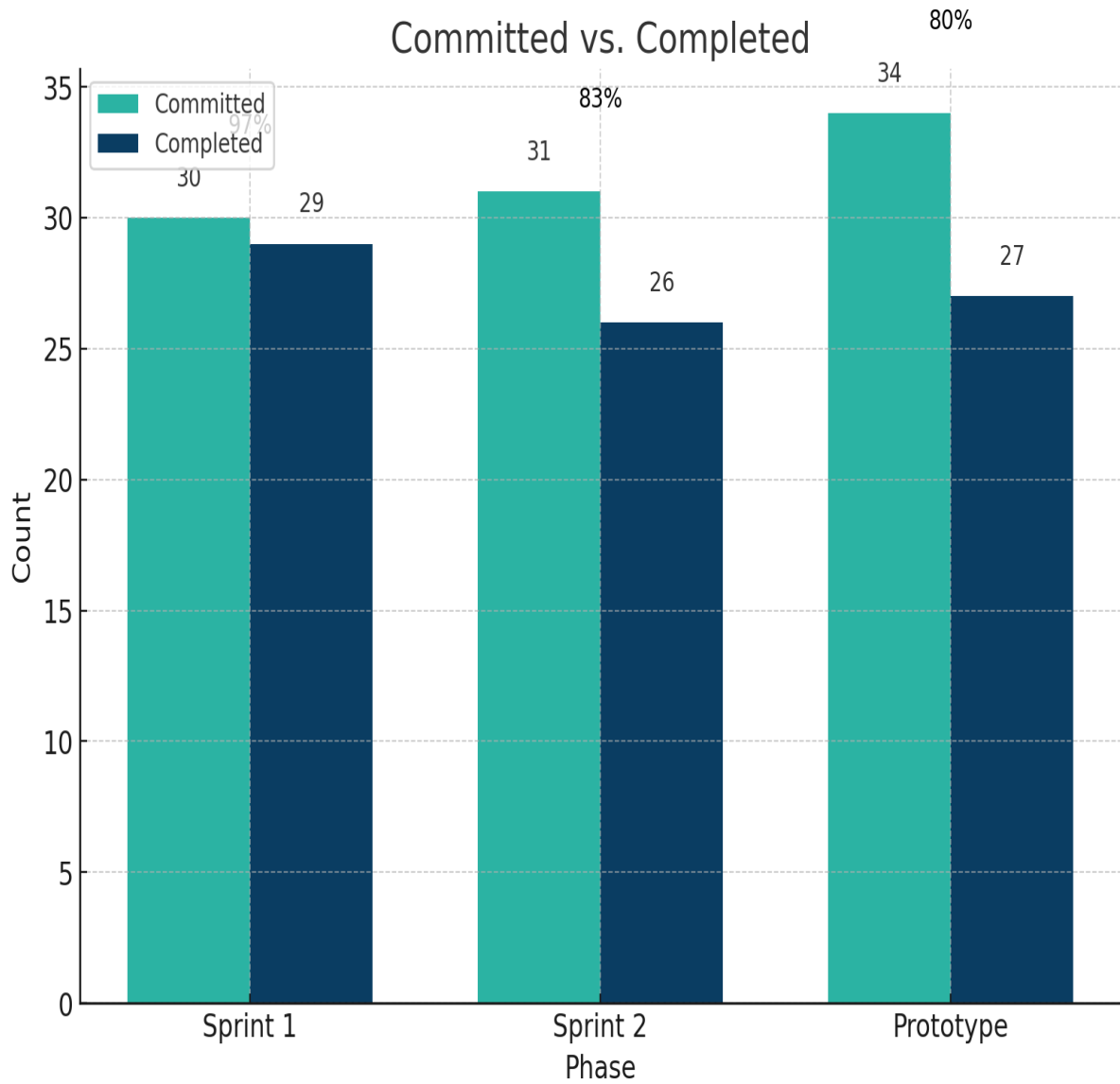


Figure 3.1 Committed Vs Completed User Stories Graph

CHAPTER 4

CONCLUSION & FUTURE ENHANCEMENTS

4.1 Conclusion

The project titled "End-to-End Application for All Construction Needs" has successfully achieved its objective of creating a centralized digital platform to streamline the process of connecting clients with verified and reputed construction service providers.

The platform enhances user experience by offering intuitive interfaces for both clients and service providers, secure authentication and authorization, and seamless service listing and discovery.

During the execution of the project:

- Core features such as user registration, profile management, builder verification, and project listing were implemented.
- Multiple sprint cycles led to iterative development and integration of key modules.
- Real-time data exchanges and asynchronous background processes were effectively handled using RESTful APIs and message queues.
- The system was tested for functionality using defined test cases to ensure robustness and reliability.
- Overall, the project stands as a scalable and secure solution for bridging the gap between users and verified construction professionals, with emphasis on trust, transparency, and accessibility.

4.2 Future Enhancements

Although the project has met its current goals, there are several potential enhancements that can be incorporated in future iterations to further enrich the platform:

- **AI-Based Builder Recommendation System:** Integrating a machine learning model to suggest the most suitable builders based on user preferences, location, budget, and past ratings.
- **Integrated Chat and Support Module:** Allowing clients to communicate with service providers or platform support in real-time.
- **Mobile Application Version:** Expanding platform accessibility by developing native mobile apps for Android and iOS users.
- **Payment Gateway Integration:** Allowing secure payments and milestone-based financial transactions within the platform.
- **Advanced Analytics Dashboard:** Providing both clients and admins with detailed insights and visual reports on performance metrics and user activity.
- **Multilingual Support:** Introducing language options to cater to a diverse user base from different regions.
- These enhancements will significantly contribute to the platform's versatility, user engagement, and long-term sustainability.

APPENDIX

A. SAMPLE CODING

```
28 -- Create users Loading...
29 CREATE TABLE IF NOT EXISTS users (
30   user_id uuid PRIMARY KEY DEFAULT gen_random_uuid(),
31   name text NOT NULL,
32   email text UNIQUE,
33   phone_number text UNIQUE,
34   user_type text NOT NULL CHECK (user_type IN ('client', 'builder', 'admin')),
35   is_verified boolean DEFAULT false,
36   verification_status text DEFAULT 'pending' CHECK (verification_status IN ('pending', 'verified', 'reject
37   profile_photo text,
38   created_at timestamptz DEFAULT now(),
39   updated_at timestamptz DEFAULT now()
40 );
41
42 -- Create builders table
43 CREATE TABLE IF NOT EXISTS builders (
44   builder_id uuid PRIMARY KEY DEFAULT gen_random_uuid(),
45   user_id uuid REFERENCES users(user_id) ON DELETE CASCADE,
46   company_name text NOT NULL,
47   license_no text UNIQUE,
48   gst_number text UNIQUE,
49   is_gst_verified boolean DEFAULT false,
50   experience_years integer,
51   category text CHECK (category IN ('newcomer', 'beginner', 'established')),
52   address text,
53   city text,
54   state text,
55   specializations text[],
56   portfolio_images text[],
57   projects_completed integer DEFAULT 0,
58   created_at timestamptz DEFAULT now(),
59   updated_at timestamptz DEFAULT now()
```

Figure 4.1 Sample Backend Code

```
{ } package-lockjson > ...
4   "lockfileVersion": 3,
5   "requires": true,
6   "packages": {
7     "": {
8       "name": "vite-react-typescript-starter",
9       "version": "0.0.0",
10      "dependencies": {
11        "lucide-react": "^0.344.0",
12        "react": "^18.3.1",
13        "react-dom": "^18.3.1",
14        "react-router-dom": "^6.22.3"
15      },
16      "devDependencies": {
17        "@eslint/js": "^9.9.1",
18        "@types/react": "^18.3.5",
19        "@types/react-dom": "^18.3.0",
20        "@vitejs/plugin-react": "^4.3.1",
21        "autoprefixer": "^10.4.18",
22        "eslint": "^9.9.1",
23        "eslint-plugin-react-hooks": "^5.1.0-rc.0",
24        "eslint-plugin-react-refresh": "^0.4.11",
25        "globals": "^15.9.0",
26        "postcss": "^8.4.35",
27        "tailwindcss": "^3.4.1",
28        "typescript": "^5.5.3",
29        "typescript-eslint": "^8.3.0",
30        "vite": "^6.2.6"
31      }
32    },
33    "node_modules/@alloc/quick-lru": {
34      "version": "5.2.0",
```

Figure 4.2 Sample Frontend Code

B. PLAGIARISM REPORT

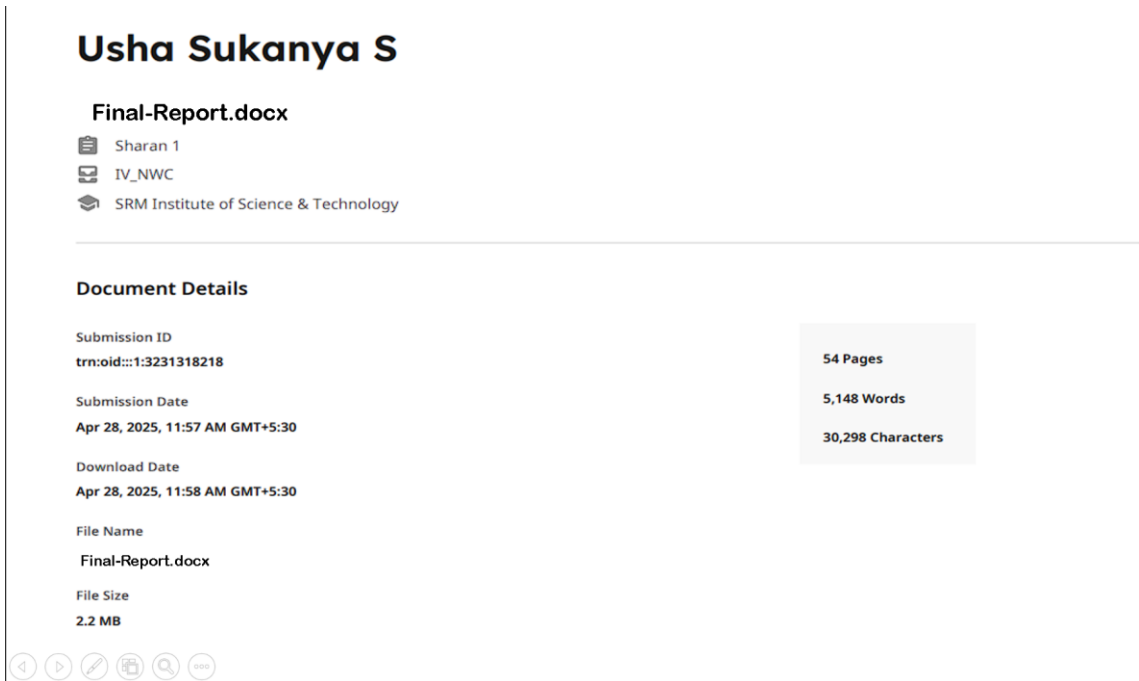


Figure 4.3 Plagiarism Report Submission

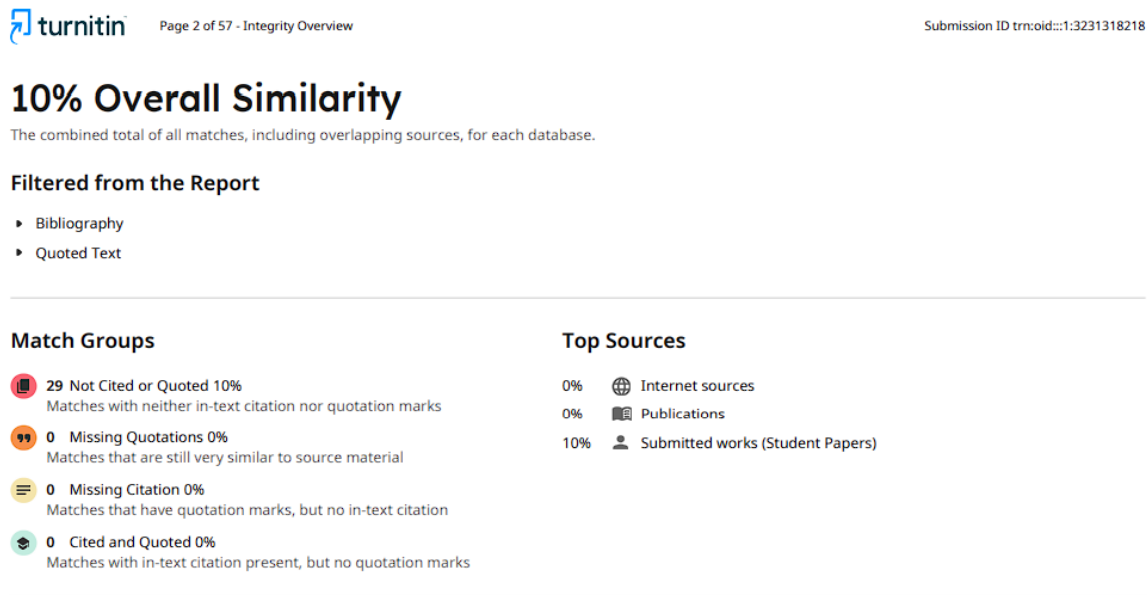


Figure 4.1 Plagiarism Report Result

PLAGIARISM REPORT

Format - I

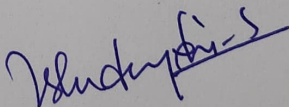
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Office of Controller of Examinations		
REPORT FOR PLAGIARISM CHECK ON THE DISSERTATION/PROJECT REPORTS FOR UG/PG PROGRAMMES (To be attached in the dissertation/ project report)		
1	Name of the Candidate (IN BLOCK LETTERS)	VEDANT PANDEY CHIRANJEEV KUMAR
2	Address of the Candidate	SANNASI-C HOSTEL
3	Registration Number	RA2211029010013 RA2211029010019
4	Date of Birth	05/06/2003 27/09/2004
5	Department	Computer Science and Engineering
6	Faculty	Engineering and Technology, School of Computing
7	Title of the Dissertation/Project	An end-to-end application for all construction needs
8	Whether the above project /dissertation is done by	Individual or group : (Strike whichever is not applicable) a) If the project/ dissertation is done in group, then how many students together completed the project : b) Mention the Name & Register number of other candidates :
9	Name and address of the Supervisor / Guide	Dr. Ushasukhanya S Mail ID: ushasuks@srmist.edu.in Mobile Number:9841669089
10	Name and address of Co-Supervisor / Co- Guide (if any)	NA Mail ID:NA Mobile Number: NA

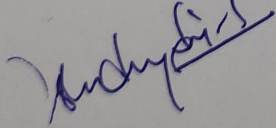
11	Software Used	Turnitin
12	Date of Verification	28/04/2025

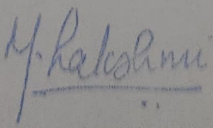
13 Plagiarism Details: (to attach the final report from the software)

Chapter	Title of the Chapter	Percentage of similarity index (including self citation)	Percentage of similarity index (Excluding self-citation)	% of plagiarism after excluding Quotes, Bibliography, etc.,
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2	Sprint Planning and Execution	1%	1%	1%
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4	Conclusion & Future Enhancements	1.04%	1.04%	1.04%
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